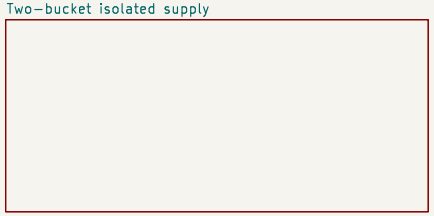
Two-bucket isolated supply



File: psu.kicad_sch

Power: 230 V AC -> IRM-05-15 (15 V; GND_C bonded to PE) -> LM317 CC 0.2 A -> LM317 CV 10.76 V -> SS34

-> two 2.5 F supercap buckets, swapped every ~15 s by four SIL05-1C90-51D reed relays (staggered, dry load break)

-> receiver side: 2200 uF -> LT3045 6.98 V -> common-mode choke -> J2 to the amplifier.

Board: 150 x 90 mm, 2 layers; mains >= 2.5 mm L-N and to PE, >= 6.4 mm to everything else (layout.py).

The only LED (D2, mains present) is on the charger side; the receiver side has none.

Buckets swing 9.4-10.0 V (CV 10.76 V, two SS34s: 2.6 V per cell); 15 s at ~0.1 A droops ~0.6 V; the LT3045 keeps its headroom in the worst corner (simulations/psu). Swap rate 33 mHz, far below the ELF band.

+9V_OUT carries the 6.98 V output (net name kept from rev 0.1).

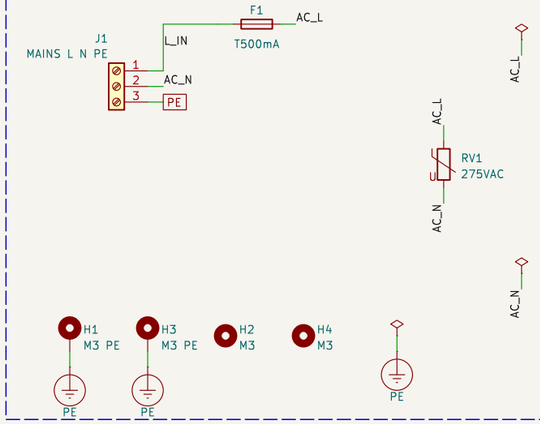
MAINS VOLTAGE on the primary side -- 6.4 mm creepage to everything else
Generated by PCB/acdc_converter/design.py -- edit the script, not the sheet
CERN-OHL-W-2.0
ELARA -- ELF Atmospheric Radio Analyser

Sheet: /
File: acdc_converter.kicad_sch

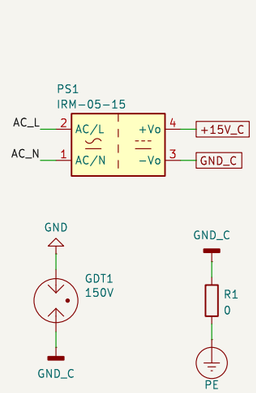
Title: ELARA PSU

Size: A4	Date: 2026-09-30	**Rev: 0.2**
KiCad E.D.A. 9.0.9		Id: 1/2

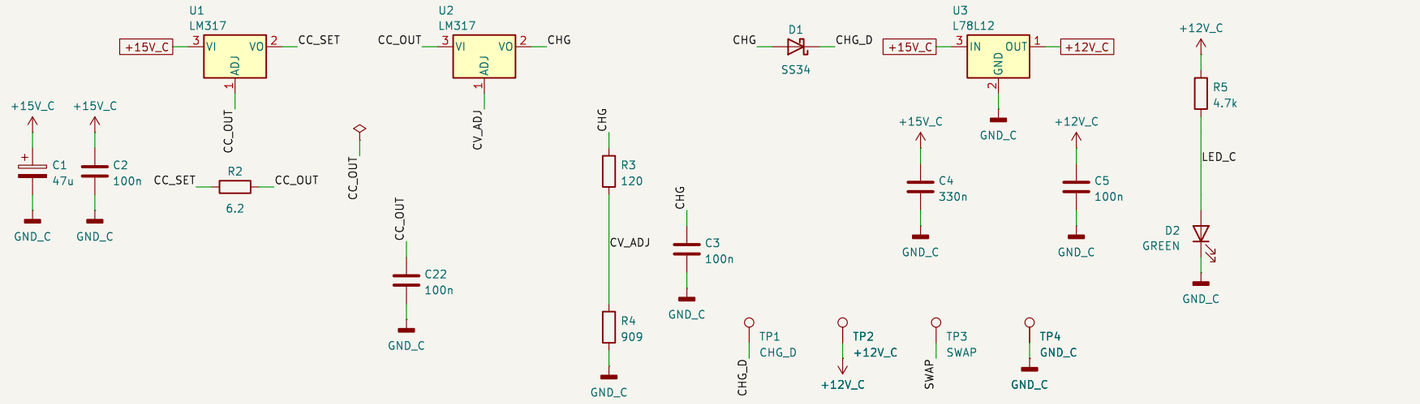
MAINS (primary)



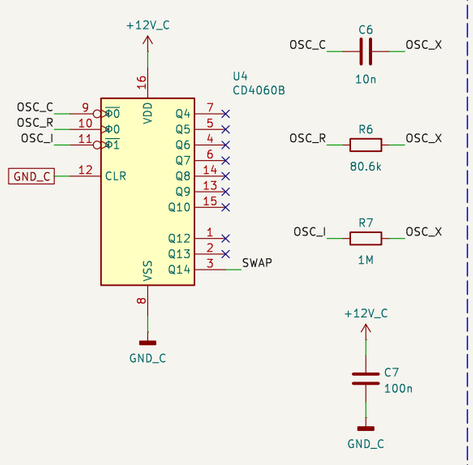
AC–DC 15 V



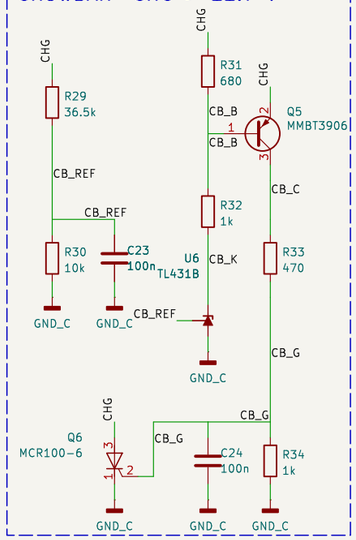
CHARGER (GND_C) CC 0.2 A -> CV 10.76 V



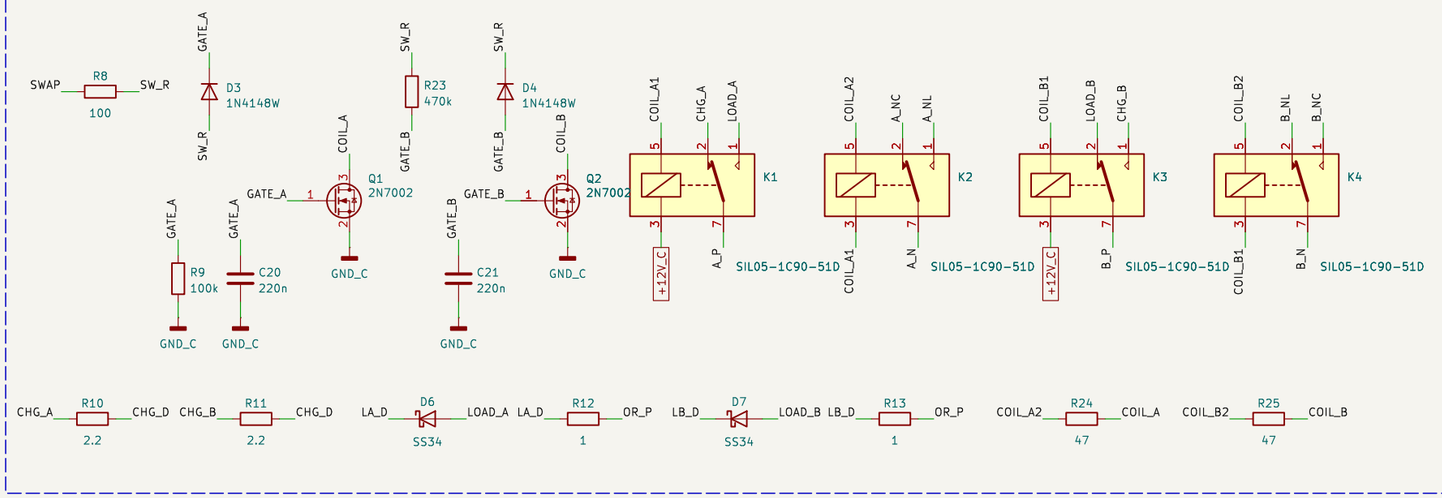
SWAP TIMER Q14 toggles every ~15 s



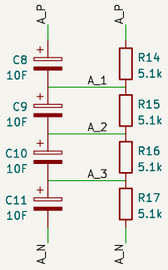
CROWBAR CHG > 11.7 V



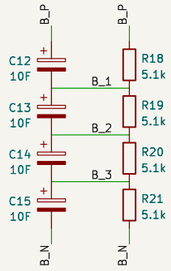
BUCKET SWAP (reed, form C; the arriving bucket switches first)



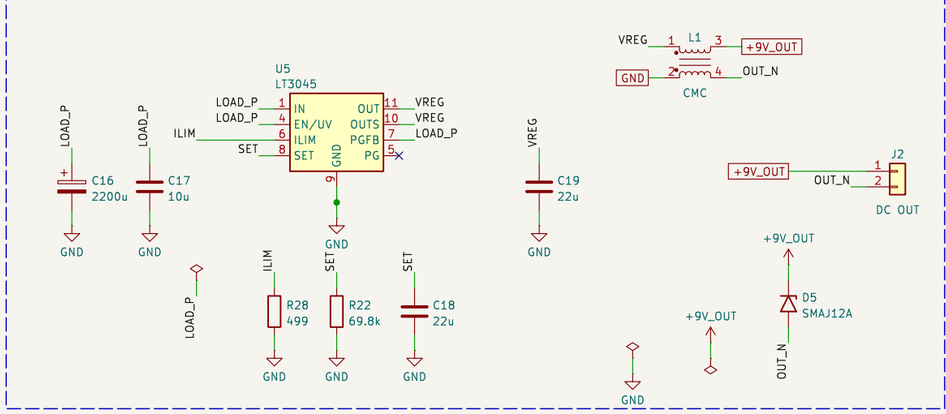
BUCKET A 2.5 F



BUCKET B 2.5 F

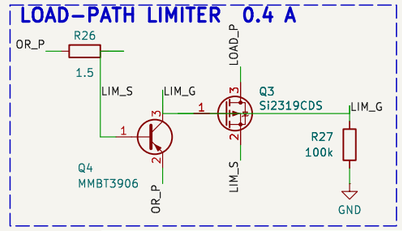


RECEIVER SIDE LT3045 6.98 V





LOAD–PATH LIMITER 0.4 A



RECEIVER TEST POINTS



MAINS VOLTAGE on the primary side -- 6.4 mm creepage to everything else
Generated by PCB/acdc_converter/design.py -- edit the script, not the sheet
CERN-OHL-W-2.0

ELARA -- ELF Atmospheric Radio Analyser

Sheet: /Two-bucket isolated supply/
File: psu.kicad_sch

Title: ELARA PSU — Two-bucket isolated supply

Size: A3	Date: 2026-09-30	**Rev: 0.2**
KiCad E.D.A. 9.0.9		Id: 2/2